Observed colour after joint stellar-mass/redshift matching
Probability-weighted classes · cells 0.05 dex × 0.005 in z

Void
Wall
Filament
Cluster

0.15 ≤ z < 0.20 · log M★ ≥ 10.60
Common support: 528,712 (99.0%); 189 cells

0.20 ≤ z < 0.22 · log M★ ≥ 10.69
Common support: 229,987 (99.1%); 75 cells

0.20 ≤ z < 0.30 · log M★ ≥ 10.96
Common support: 601,139 (98.4%); 262 cells

Before weighting
Common-support density [mag⁻¹]

Cluster − void mean colour
+0.0312 [+0.0289, +0.0333] mag

Cluster − void mean colour
+0.0317 [+0.0286, +0.0344] mag

Cluster − void mean colour
+0.0312 [+0.0294, +0.0328] mag

Mass/redshift matched
Density [mag⁻¹]

Matched − pooled matched
Density difference [mag⁻¹]

Observed g − r [mag]

Same retained galaxies before/after weighting; all four environments share the pooled joint cell distribution.
Bands: pointwise conditional 95% sky-block bootstrap, refitting weights with fixed cells/target. Pooled curve: equal mean of four classes.
All finite colours enter normalization; display zoom 0.3–1.9 mag. Observed colours/CIGALE masses; no causal or calibration claim.